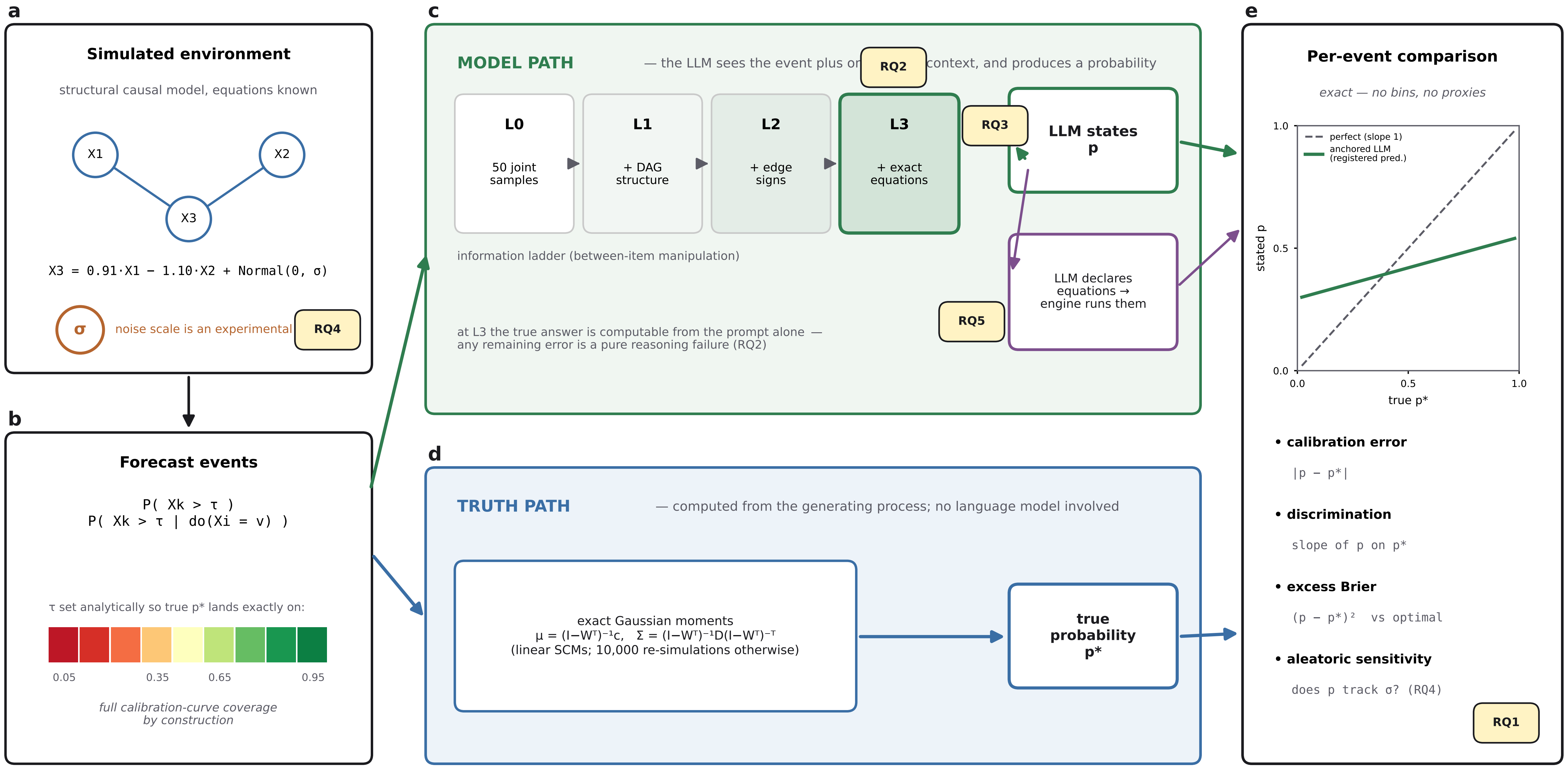
Measuring LLM calibration against knowable probabilities



Study design.

Each forecast event is posed about a simulated system whose generating equations are known, with thresholds placed so the true probability p* spans the full unit interval. The same event travels two paths: the model path (the LLM answers, given one rung of the information ladder — or declares its own equations, which the engine executes), and the truth path (p* computed exactly from the mechanism). Comparing p to p* per event yields calibration, discrimination, and distance-from-optimal without bins or proxies — and dialing the noise σ tests whether stated uncertainty tracks true randomness. RQ1-RQ5 mark where each research question is answered.